

# Understand Equations and Their Solutions

Lesson 8-1

**Name:**

**Date:**

**Class:**

COLD CASE MISSION

## | The Equation Vault

You are Detective Ruiz's new code analyst. A tip just came in: 'A number of stolen gems plus 8 more equals 20 total.' The evidence vault only opens when the clue is translated into a correct equation. Master writing equations and the case cracks wide open.

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can write an equation for a real situation and check whether a given value is a solution by substituting it.

**Language Objective:** I can explain a solution using the words substitute, solution, true, and equal sign.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Write Equations

Equation

Variable

Equal sign

Expression

Constant

Unknown

Solution

Substitute



Tap any word to see what it means and a picture.

1

Two expressions joined by an   
 make an equation.

2

A math sentence that says two amounts are equal, with an = sign, is an  
.

3

A letter that stands for an unknown number is a .

4

The symbol = that shows two amounts are the same is the  
.

5

A group of numbers and variables with no equal sign is an  
.

6 A fixed number that does not change is a .

7 The value you are trying to find in an equation is the .

8 A value that makes the equation true when  for the variable is its solution.

9 Replace the variable with a number to  whether it is a solution.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How many paintings has the detective recovered? How did writing an equation help you understand the situation?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Write Equations** — Represent a relationship with expressions joined by an equal sign.

*Escribir ecuaciones — Representar una relación con expresiones unidas por un signo igual.*

- ☐ **Equation** — A math sentence with an equal sign showing both sides are the same.

*Ecuación — Una oración matemática con un signo igual que muestra que ambos lados son iguales.*

- ☐ **Variable** — A letter that stands for a number that is unknown or can change.

*Variable — Una letra que representa un número desconocido o que puede cambiar.*

- ☐ **Equal sign** — The = sign, showing both sides are the same.

*Signo igual — El signo =, que muestra que ambos lados son iguales.*

- ☐ **Expression** — A math phrase with numbers and letters, but no equal sign.

*Expresión — Una frase matemática con números y letras, pero sin signo igual.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The detective recovered \_\_\_\_ paintings because  $p + 14 = 37$  means  $p = 37 -$   
\_\_\_\_ = \_\_\_\_.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** I let a variable stand for the unknown number of gems.

**Evidence:** Students choose a variable (such as  $n$ ) for the unknown gems, build  $n + 8 = 20$ , and connect 'plus 8 more' to  $+ 8$  and 'equals 20 total' to  $= 20$ .

**Reasoning:** This evidence connects the definition of equation to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The detective recovered \_\_\_\_ paintings because  $p + 14 = 37$  means  $p = 37 - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ .
- My answer is \_\_\_\_.

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.
- Mi afirmación es \_\_\_\_.*
- I know because \_\_\_\_.

*Lo sé porque \_\_\_\_.*

- So \_\_\_\_.

*Entonces \_\_\_\_.*

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#### Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**

*Mi afirmación es \_\_\_\_.*

- **My math evidence is \_\_\_\_.**

*Mi evidencia matemática es \_\_\_\_.*

- **This proves \_\_\_\_ because \_\_\_\_.**

*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

☐

I answered the question.

*Respondí la pregunta.*

☐

I used math evidence: numbers, an equation, a model, or a comparison.

*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

☐

I explained how the evidence proves my answer.

*Expliqué cómo la evidencia demuestra mi respuesta.*

☐

I used at least two math words.

*Usé por lo menos dos palabras matemáticas.*

☐

I reread complete sentences and punctuation.

*Volví a leer mis oraciones completas y la puntuación.*